

# Eloquent lab proposal for CLEF 2024

Jussi Karlgren, Aarne Talman, Liane Guillou, Joakim Nivre,  
Luise Dürlich, Evangelia Gogoulou, Magnus Sahlgren

Large language models (LLMs) have become a foundational technology for both research and information service development in the past year, and have prompted great public interest through some quite innovative application scenarios. Large language models as a tool are much in evidence in academic research, not least in many of the CLEF labs.

New models with generative capacity are released with very various capabilities and intended usage scenarios. The practical application of such models in public arenas, commercial and elsewhere, is challenged by run time costs, training costs, and to a great extent by quality concerns about predictability and reliability of the output from a generative component. A language model can provide a system with fluently generated output but without guarantees of relevance, veracity, or consistency. Quality assurance, evaluation, and benchmarking of large language models is performed using tests that focus on details of language understanding and reasoning, but many factors that are crucial for application development and deployment to third parties are not addressed systematically through existing benchmarks.

This proposed evaluation lab is intended to experiment with new evaluation methods for generative language models to meet some of the challenges in the path from laboratory to application. The organisers include commercially active AI developers as well as research groups, and our intention is to propose a series of tasks to be developed iteratively over three years, each year drawing from experiences from the previous year. In a departure from CLEF practice, we intend to develop tasks which do not rely on gold standards and which involve minimal human effort expended on assessment, instead leveraging the intrinsic qualities of generative AI for test development and scoring.

The outcome of this proposed lab is not only to score the performance of participants' submitted models and to provide benchmarks for future reuse but to more generally point out future directions for evaluation of generative and learning models, following up on the 2015 CLEF paper on "Evaluating Learning Language Representations".

## I. USAGE SCENARIOS AND DOMAIN

This type of evaluation will find its natural home at CLEF: Europe is a truly multilingual arena, with a large number of equally relevant languages and language varieties, dialects, and sociolects, none of which can be disregarded in commercial or public applications of information technology. A vibrant technology scene provides great readiness for deploying tools and services to a multi-lingual and increasingly mobile population with a high level of acceptance to information services. To provide inclusive services to a diverse population will require

a large variety of background models with verified equivalent or comparable competence in several languages and language varieties.

Many practical application scenarios for generative AI require systems and models that are severely constrained by time or energy budget. This requires building parsimonious models using larger models as a guide which in turn requires a systematic verification of the quality of tailored, tuned, distilled, or extracted models compared to large all-round models.

The motivation for using a generative language model as a component in an information service is to handle a broad variety of input data robustly and elegantly, and to provide appropriately generated output to fit the varied application situations and the needs of a diverse user population. A classic gold standard-based evaluation may provide a useful sampling of such conditions, but validating this capability is better done in realistic situations which provide varied and unpredictable input as well as varied and unpredictable requirements for output.

Organisations, whether commercial or not, will likely be unable to expend great human effort on testing, benchmarking, sampling, and validation to perform quality assurance on a language module which oftentimes is a subsidiary component of a larger envisioned service. Effective testing will therefore require automated or semi-automated testing procedures, and leveraging the capabilities of language models to self-assess its qualities.

## II. ORGANISERS

This lab is organised by Silo AI, AI Sweden, and RISE Research Institutes of Sweden. Silo AI will be responsible for the overall coordination of the lab. All organisers will contribute to the task formulation and resource development. Silo AI will chair the Topical Test Generation task; AI Sweden will chair the Robustness task; RISE will chair the Veracity task. In addition we propose the Voight-Kampff task, to be organised together with the PAN lab.

**Silo AI** is one of Europe's largest private AI labs with customers in fields ranging from smart devices, autonomous vehicles, industry 4.0, to smart cities. Established in 2017, Silo AI is on a mission to build a European flagship AI company, with offices currently in Finland, Sweden, Denmark, Switzerland, the Netherlands, and Canada. Silo AI is currently working on making generative AI an industrially reliable and verifiably practical solution for its customers. This initiative will contribute to that effort and Silo AI will be able to validate the evaluation approaches and offer a path to industrial application of participants' contributions.

- **Jussi Karlgren** has participated in every CLEF. He was a member of the CLEF Steering Committee from CLEF's foundation to 2009. He has been the co-organiser of iCLEF, the interactive search lab at CLEF (2006-2009), and the Podcast Track at TREC (2020-2021).
- **Aarne Talman** is Head of Technology for SiloGen at Silo AI and Visiting Scholar in Language Technology at University of Helsinki. He has over 16 years of industry experience ranging from software development to technology leadership and academic research.

**AI Sweden** is the national centre for applied AI in Sweden, which brings together more than 100 partners across the public and private sectors as well as academia. AI Sweden is funded by the Swedish government through the Swedish innovation agency (Vinnova), and by partners, both public and private. Together, we invest in generating tools and resources to accelerate the use of AI for the benefit of our society, our competitiveness, and everyone living in Sweden. AI Sweden has three main focus areas: AI transformation and adoption, ecosystem and relations, and AI labs in three different areas: NLU, data factory, and edge learning. The NLU research group focuses specifically on large language models, and has developed the first large language model for the Nordic languages, GPT-SW3.

- **Magnus Sahlgren** is Head of research for NLU at AI Sweden. He has a PhD in computational linguistics, and has worked on computational models of meaning for more than 20 years in both research and industry. Sahlgren has participated in several CLEF campaigns, and has organised several international workshops.

**RISE Research Institutes of Sweden** is a national research institute, whose department of computer science (and its predecessor, the Swedish Institute of Computer Science, SICS) has a long tradition of research in natural language processing and information retrieval, including groundbreaking work in distributional semantics. The NLP group at RISE has recently been involved in efforts to train large generative language models for Scandinavian languages, in collaboration with AI Sweden. Other prominent research themes are multilingual language models and resource-efficient NLP. Several researchers at RISE have extensive experience of organising shared tasks in NLP.

- **Liane Guillou** is a postdoctoral researcher at The University of Edinburgh, and will be a researcher at RISE from August 2023. The central themes of her research include evaluation, multilinguality, and discourse. Liane was the lead organiser for the shared task on Cross-lingual Pronoun Prediction at the Conference on Machine Translation (WMT) in 2016, and a co-organiser for the pronoun test suite evaluation at WMT 2018. Most recently, she participated in the challenge set subtask of the metrics shared task at WMT 2022.
- **Luise Dürlich** is a PhD student at Uppsala University, RISE and the Swedish Medical Products Agency with an interest in explainability, resource efficiency, and multilinguality.

- **Evangelia Gogoulou** is a PhD student at KTH in Stockholm and at RISE with a research interest in contextualised language models.
- **Joakim Nivre** is senior researcher at RISE and professor of computational linguistics at Uppsala University. He is one of the main developers of the transition-based approach to syntactic dependency parsing and a co-founder of the Universal Dependencies project, which develops annotated corpora for the world's languages and currently involves over 500 researchers around the world. He has extensive experience of organising shared tasks, in particular at the Conference on Computational Natural Language Learning (CoNLL), where he was responsible for shared tasks in 2007, 2008, 2009, 2017 and 2018. He is a fellow of the Association for Computational Linguistics and was the president of the association in 2017.

A steering committee for the lab will be constituted after the lab has been announced at CLEF 2023, with members selected among interested participants, to represent the breadth of interests we expect to find. New members for the steering committee will be selected each year.

### III. FORMAT AND SESSION

This lab will be run as an experimental track. We propose to run the task with a three-year perspective, with the first year more exploratory and the two following years in a more consolidated manner. We expect the presentation session at CLEF 2023 to be quite lively and are willing to accommodate suggestions from potential participants to fine-tune the tasks for both feasibility and external impact. We propose to hold the session at CLEF 2024 in a half day but would like to reserve the option to increase it if participation exceeds our expectations. This will be evident early in 2024.

We expect the session to be run in well-established CLEF and TREC style with a high-level overview of results and with most interesting (not necessarily the most highly scoring) participants invited to present their approaches in brief. We also plan that in the first year a fair slice of the time will be allocated to discuss the task structure and propose modifications and improvements to them.

### IV. AUDIENCE

The organisers have well established and extensive networks in academic research, industrial research, public policy, and commercial application of large language models and generative AI technology. Many CLEF participants are already working with large language models in the course of their experimentation with other evaluation labs and participating in this proposed task will require comparatively little marginal effort for them; the organisers have among them direct links to most research groups on LLM development in Europe.

## V. TASKS

The tasks are planned to involve comparatively little human effort, and instead leverage the capacity of the current generation of AI models to generate, process, and assess human language input. For this first year we propose four tasks, one of which will be co-organised with the PAN lab, and one other where we are exploring a collaboration with the CheckThat! lab. One task — the topical test generation task — is intended to have a low threshold for participants to sign up; the other tasks will involve more complexity. We will poll the interest among participants before we nail down the details of the proposed tasks.

The organisers wish to encourage efforts in multilingual research and as such do not intend to place any restrictions on the languages that participants may work with. For tasks in which interaction between models is required, models developed for the same language may be paired together. In cases where this is not possible, the application of automated translation will be considered as a bridging mechanism. Some of these design choices will be discussed among participants before the final version of the tasks is launched.

- 1) **Topical Test Generation Task:** The task is to generate a topical competence test on a set of given topics in the form of a number of topically relevant questions. These questions will then be posed to other models participating in the task in a round-robin fashion, with the responses assessed by the model that generated the questions. Quality of the test will be assessed by cross-model correlation of test results. This task answers to the need for verifying a model's understanding of an application domain of interest and will also probe the capacity of a model to communicate with another model. It is our intention, once this task has been executed in the first year, to evolve it into a test for *Conversational competence*, where two models or two instances of a model will be set up to engage in a conversation to solve some given task which will involve making use of both topical and conversational competence.
- 2) **Veracity Detection Task:** Detecting hallucinated or factually incorrect information may be difficult for human users of LLMs or LLM-based systems and systems should be rigorously tested for their ability to generate accurate output prior to deployment. Inspired by the *Build it, break it: Language Edition* task<sup>1</sup> and the *WMT metrics shared task* challenge set subtask<sup>2</sup>, we divide the task into two subtasks: a builders subtask which focuses on model development, and a breakers subtask which focuses on evaluation. We are in a conversation with the CheckThat! lab organisers about how to execute this task jointly.
  - A (builder): participants develop models – baselines or models designed to be truthful. Within a Question Answering setting, models will evaluate the veracity of answers output by other models (or another instance of itself) with appropriate prompts used to a) generate true/false answers to a question, and b) evaluate an answer made by another model. Models will be assessed using a small ground truth dataset, and the datasets from subtask B. This task will assess the viability of cross-model and self evaluation.
  - B (breaker): participants construct novel evaluation benchmarks. The aim is for these datasets to be of human or near-human quality – e.g. participants might use a range of pre-existing data resources and apply automated techniques such as data augmentation (e.g. using LLMs). This task will explore the use of automated methods for generating high-quality evaluation benchmarks. The output of this task in the first year will be pooled and provide guidelines for subsequent cycles' development.
- 3) **Robustness Task:** The task is to test the capability of LLMs to handle dialectal, sociolectal, and cross-cultural variation as represented by human-generated varieties of input prompts. The results will be assessed by how variation in output is conditioned on variation of equivalent but non-identical input prompts. We expect to execute this task in a builder-breaker style similar to the Veracity Detection task.
- 4) **Voight-Kampff Task:** Recent advances in LLMs have made it possible to automatically generate content for websites, news articles, social media, etc. Labelling AI-generated content, in order to tackle disinformation, is something that the EU has recently urged technology companies to do.<sup>3</sup>

In this task we aim to discover whether automatically-generated text can be distinguished from human-authored text. Detecting automatically generated text, with the increased quality of generative AI, is becoming a task quite similar to human authorship verification, for which we can build on years of experience from previous shared tasks organised by the PAN lab, and this task will be organised in collaboration with the PAN lab at CLEF. As the previous two tasks, it will also be executed in builder-breaker style.

  - A (builder): PAN lab participants will build systems to discriminate between automatically-generated and human-authored texts. The systems will be evaluated using the dataset constructed in subtask B.
  - B (breaker): ELOQUENT lab participants will focus on the construction of the evaluation dataset. A human-authored test set will be selected from existing texts of different styles, and the automated set will be generated by first automatically summarising the human-authored texts, and then requesting a LLM to generate a text of equal length to the original based on the summary. The LLM texts may be designed to be similar in style to the human-authored texts to avoid detection by existing LLM-detection systems.

<sup>1</sup><https://bibinlp.umiacs.umd.edu/sharedtask.html>

<sup>2</sup><https://wmt-metrics-task.github.io/>

<sup>3</sup><https://www.theguardian.com/technology/2023/jun/05/google-and-facebook-urged-by-eu-to-label-ai-generated-content>